Bibliography Report – Multimodal Fusion of sEMG, Inertial, Gaze and Vision Data for Limb-Position-Robust Grasp Type Classification in Transradial Myoelectric Prostheses

Nov 7, 2026 · @Narean R

Technische Hochschule Deggendorf – Master Robotics

Course: MRO-AR2 – Case Study: Assistive Robotics for Improvement of Life Quality (WS 2026/27)

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Project track: Software & Simulation

Portfolio component: Literature Review (Bibliographie), final version

Abstract

Myoelectric prostheses are the most common powered replacement for a lost hand, yet many users abandon them because control is unreliable in daily life. Grasp type classification from surface electromyography (sEMG) reaches high accuracy in the laboratory, where the arm is held in one posture, but its error rises several-fold as soon as the arm is used in a different position. This degradation is called the *limb position effect*, and it is one of the main reasons why laboratory results do not carry over to clinical use.

This report reviews 24 sources that together explain the problem and point towards a solution. The review follows one continuous line of reasoning. It starts with the clinical need and the abandonment of upper-limb prostheses, then describes how sEMG pattern recognition works and why it fails when the limb position changes. It then examines three additional sources of information that do not depend on muscle signals in the same way: inertial sensing, which measures the posture of the arm; computer vision from a scene camera, which suggests a grasp from the shape of the object; and eye-tracking gaze, which reveals which object the user intends to grasp before the hand reaches it. Finally, it reviews how these modalities have been fused and which public datasets and metrics allow a reproducible evaluation.

The literature shows that each modality corrects a different weakness. Inertial data make position-aware classification possible and reduced the inter-position error of transradial amputees from 29.9% to about 9%. Gaze combined with vision raised the grasp classification accuracy of amputees by about 15 percentage points. However, no reviewed study fuses sEMG, inertial, gaze and vision data while explicitly measuring robustness across limb positions, and few report what happens when one modality is missing. These gaps lead directly to our topic: a reproducible, offline simulation of multimodal fusion for limb-position-robust grasp type classification in transradial myoelectric prostheses, evaluated on the public MeganePro dataset.

Keywords: transradial myoelectric prosthesis; surface electromyography (sEMG); limb position effect; grasp type classification; inertial measurement unit (IMU); eye tracking; computer vision; multimodal sensor fusion; decision-level fusion.

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1 Introduction

1.1 Clinical motivation: upper-limb loss and prosthesis abandonment

The loss of a hand is one of the most disabling injuries a person can experience, because the hand is the main tool through which people act on their environment. Ziegler-Graham et al. estimated that about 1.6 million people were living with limb loss in the United States in 2005 and projected that this number would rise to about 3.6 million by 2050 [1]. Even though upper-limb loss is less frequent than lower-limb loss, its effect on independence is severe: eating, dressing, personal hygiene, opening doors and most kinds of work all depend on the ability to grasp and release objects. For a transradial (below-elbow) amputee, restoring reliable grasping is therefore the single most important goal of a prosthesis, and it is directly connected to the course theme of improving quality of life through assistive robotics.

Myoelectric prostheses address this need by reading the electrical activity of the residual forearm muscles with surface electrodes and converting it into commands for a motorised hand. Modern multi-articulated hands can physically perform many different grasp types, such as a power grasp around a bottle, a precision pinch for a coin or a lateral grasp for a key. The hardware, however, is only useful if the user can choose the right grasp at the right moment. Biddiss and Chau reviewed 25 years of surveys on upper-limb prosthesis use and found high rejection and abandonment rates, with poor function, difficult control and a lack of perceived benefit among the most frequently reported reasons [2]. This finding is the starting point of our work: the limitation of current myoelectric hands lies less in their mechanics than in the reliability of their control.

1.2 From pattern recognition to the limb position effect

The main research answer to this control problem is myoelectric pattern recognition. Instead of mapping one muscle to one function, a classifier learns which spatial and temporal pattern of sEMG activity corresponds to which intended movement. Hudgins,

Parker and Scott introduced this approach with a small set of time-domain features and a classifier that distinguishes several movement classes [3]. Since then, laboratory studies have regularly reported classification accuracies above 90% for able-bodied subjects and amputees.

Scheme and Englehart, in their review of the clinical state of the art, explained why this laboratory success has not translated into everyday use [4]. A classifier is trained under controlled conditions, but in daily life the sEMG signal changes for reasons that have nothing to do with the user's intention. Among the disturbances they identified, such as electrode shift, sweat, fatigue and variation in contraction force, one stands out because it occurs in almost every functional task: the position of the arm. When a user reaches for a cup on a high shelf, the forearm muscles must also hold the arm against gravity, and the geometry between muscle and electrode changes. The sEMG pattern for the same grasp therefore looks different in different arm positions. Fougner et al. showed that the classification error of able-bodied subjects rose from about 3.8% to about 18% when the classifier was used in positions other than the training position [5]. Geng, Zhou and Li confirmed the effect in transradial amputees, where the error rose from 7.3% in the training position to 29.9% in other positions [6]. This degradation is known as the *limb position effect*.

The same studies also indicated a remedy. Both Fougner et al. [5] and Geng et al. [6] used accelerometer data to estimate the arm position and then applied a position-specific classifier, which reduced the error considerably. This observation is important for our topic, because it shows that the limb position effect is not an unsolvable property of sEMG but a lack of information about the context in which the signal is produced. If the system knows the posture of the arm, and ideally also the object the user wants to grasp, it can interpret the sEMG signal more reliably.

1.3 Research question and objectives

Based on this reasoning, the research question of this project is: **how can additional sensing modalities be fused with sEMG so that a transradial myoelectric prosthesis classifies the intended grasp type reliably when the limb is in different positions?** We argue that sEMG alone cannot answer this question, because the information it lacks, namely the posture of the arm, the identity of the target object and the moment of the user's intention, is carried by other signals. Inertial sensors measure the posture, a scene camera sees the objects in front of the user, and an eye tracker shows which of these objects the user is looking at.

The literature review pursues four objectives that build on each other. The first is to establish the state of the art in sEMG pattern recognition and to quantify the limb position effect from published results. The second is to examine how inertial, vision and gaze data have been combined with sEMG and what each modality contributes. The third

is to identify public datasets and evaluation metrics that allow a reproducible, simulation-based study within one semester. The fourth is to derive from the identified research gaps a specific topic with clearly defined inputs, system, outputs, requirements and target values.

1.4 Structure of the report

Chapter 2 defines every key term of the project title, so that each word has one meaning and one measurable criterion throughout the portfolio. Chapter 3 describes how the literature was searched and selected, and how the review narrowed the broad course theme to our specific topic. Chapter 4 presents the literature as one continuous argument, moving from sEMG control to the limb position effect and then to the three complementary modalities and their fusion. Chapter 5 compares the works, states the research gaps and shows how each gap shaped a part of the topic. Chapter 6 outlines the proposed system and method, and Chapter 7 concludes the report. References are numbered in the order in which they are first cited.

2 Key Terms and Definitions

The title of this project is: *Multimodal Fusion of sEMG, Inertial, Gaze and Vision Data for Limb-Position-Robust Grasp Type Classification in Transradial Myoelectric Prostheses.* Every word in this title was chosen to match the terminology of the reviewed literature, so that the topic can be searched, compared and assessed against existing work. To avoid ambiguity, each term is used with exactly one meaning in this report and in all later portfolio components. In particular, we consistently write *grasp* rather than *grip*, and *limb position* rather than *arm position* when referring to the effect, because these are the terms used in the core sources [5], [6]. Table 1 gives the definition of each term and the criterion by which it will be measured in our project.

Table 1: Key terms of the project title, their definitions and how each is measured.

Term	Definition used in this project	How it is measured or shown
Multimodal	Using signals of physically different types: electrical muscle activity (sEMG), inertial (acceleration), gaze (eye movement) and visual (scene images)	Four modalities, each with its own processing branch

Term	Definition used in this project	How it is measured or shown
Fusion	Combining the information of several modalities into one decision; here a hybrid of a hierarchical cascade (inertial data select the sEMG model) and decision-level fusion with gaze and vision	Ablation study: accuracy with each modality removed in turn
sEMG	Surface electromyography: electrical activity of the residual forearm muscles recorded with skin electrodes	12 channels in the selected dataset
Inertial data	Accelerometer signals from sensors on the forearm, used to estimate the orientation of the limb	Limb position labels derived from accelerometer orientation
Gaze data	The point the user is looking at, recorded by a head-mounted eye tracker	Fixations on objects before grasp onset
Vision data	Video from a head-mounted scene camera showing the objects in front of the user	Objects detected and linked to compatible grasp types
Limb position	The posture of the arm during reaching and grasping, for example arm down, arm forward or arm raised	Discrete position classes from the accelerometer data
Limb position effect	The increase in sEMG classification error when the classifier is used in a limb position different from the training position	Error difference between same-position and different-position tests
Robust	The grasp classification accuracy stays high when the limb position changes	Inter-position accuracy drop compared with the sEMG-only baseline
Grasp type classification	Choosing one discrete grasp type (for example power, pinch or lateral) out of a fixed set, plus rest; not the control of grasp force or of single fingers	Classification accuracy (%) per subject

Term	Definition used in this project	How it is measured or shown
Transradial	Amputation below the elbow, where forearm muscles remain available for sEMG	Results reported separately for amputee subjects
Myoelectric prosthesis	A powered prosthetic hand controlled by sEMG signals	Offline simulation of the controller; no physical prosthesis

Two boundaries follow from these definitions. First, *classification* means that the system outputs one discrete grasp type before the hand closes; continuous control of force or finger positions is outside the scope. Second, *robust* is never used without a metric: in this project it always refers to the measured accuracy drop between the training position and other limb positions. In the same way, words such as *better* or *efficient* are only used together with a number and a reference.

3 Review Methodology

3.1 Search strategy and keywords

The literature was searched in IEEE Xplore, PubMed and PubMed Central, ScienceDirect, Google Scholar, the Frontiers journals and Nature Scientific Data. These databases were chosen because together they cover biomedical engineering, rehabilitation research, robotics and open data publications. The search was carried out in stages that mirror the structure of the review. We first searched for the clinical problem with the terms *upper-limb prosthesis*, *prosthesis abandonment* and *myoelectric control*. Once the limb position effect had been identified as the central obstacle, we searched specifically for *limb position effect*, *arm position*, *forearm orientation* and *EMG pattern recognition*. In the third stage, we searched for complementary modalities with the terms *inertial measurement unit*, *accelerometer*, *computer vision grasp*, *eye tracking*, *gaze* and *visuomotor*, each combined with *prosthesis* or *sEMG* using AND. The last stage focused on combination and evaluation, with the terms *sensor fusion*, *multimodal*, *semi-autonomous control*, *grasp taxonomy*, *dataset* and *transradial amputee*.

Working in stages meant that each set of results determined the keywords of the next search. For example, the remedy proposed by Fougner et al. [5] and Geng et al. [6], an accelerometer-based estimate of arm position, is what led us to search for inertial sensing as a separate modality. In the same way, the weakness of vision in scenes with several objects led us to the literature on gaze. The keywords listed in the abstract are the final, stable set that resulted from this process.

3.2 Selection criteria and screening

A source was included when it was a peer-reviewed journal article, a conference paper or a published dataset; when it used non-invasive sensing on the forearm, the head or the scene; and when it reported a quantitative result, such as accuracy, error rate or task completion rate, or provided open data. Foundational papers were included regardless of their age, because the basic methods of sEMG pattern recognition and the basic findings on eye-hand coordination have not changed. Application papers were mainly selected from 2011 onwards, the year in which the limb position effect was systematically quantified. Sources were excluded when they dealt with invasive interfaces such as implanted electrodes or targeted muscle reinnervation, with lower-limb prostheses, or with methods that were not evaluated.

Screening was done in two passes. Titles and abstracts were read first to remove clearly unrelated work, and the remaining papers were then read in full. For every source that was kept, we wrote a short summary in our own words that recorded its method, its main quantitative result, its limitation and its relevance to our project. This process resulted in 24 sources, consisting of 2 clinical studies, 18 methodological and experimental studies, 2 studies on human eye-hand coordination and 2 public datasets. The same summaries form the basis of the comparison in Chapter 5.

3.3 How the literature narrowed the topic

The review was not only a collection of sources but also the process through which our topic was defined. The course theme, assistive robotics for the improvement of life quality, is broad. Our earlier topic ideas, first an automated external defibrillator delivery drone and then a sensor-fusion-based tremor suppression wrist exoskeleton, were relevant to quality of life, but supervisor feedback showed that they were not yet specific enough: a topic must name one concrete problem with clearly defined inputs, system, outputs, targets and requirements, using the terms of the literature.

We therefore used the literature to narrow the theme in four steps. The course list of project directions included wearable intention detection from IMU and EMG signals, which pointed us to upper-limb prostheses as an assistive device with a well-documented user need [1], [2]. Within this field, the review of clinical challenges [4] identified unreliable control in daily life as the core problem, and the quantitative studies on arm position [5], [6] turned this general problem into a specific, measurable one: the limb position effect. Finally, the remedies proposed in these studies, together with the literature on vision and gaze reviewed in Chapter 4, defined the solution space of multimodal fusion. Each step replaced a broad word with a precise term from the literature, which is how the final title in Chapter 2 was formed.

4 Literature Review

This chapter presents the reviewed literature as one continuous argument. Each section answers a question raised by the previous one: how sEMG control works, why it fails when the limb moves, and which additional information can correct that failure. The chapter ends with the datasets and metrics that make the resulting idea testable.

4.1 sEMG pattern-recognition control

The foundation of modern myoelectric control was laid by Hudgins, Parker and Scott [3]. They divided the sEMG signal into short analysis windows and extracted four simple time-domain features from each window: the mean absolute value, the number of zero crossings, the number of slope sign changes and the waveform length. A classifier then mapped these features to a set of movement classes. The method was simple, but it showed for the first time that several hand functions could be controlled from the same set of electrodes. These features are still known as the Hudgins features and remain the standard baseline in the field.

Englehart and Hudgins extended this work into a control scheme that runs continuously in real time [7]. They used overlapping analysis windows, so that a new decision is produced at short intervals, and a linear discriminant analysis (LDA) classifier whose output was smoothed by majority voting over successive decisions. LDA has remained popular in prosthetics research and in commercial systems because it is fast to train, needs little computation and is not prone to overfitting. The real-time setting, however, introduced a new constraint: every decision must be made quickly enough for the user to feel in control. Smith et al. investigated this trade-off directly [8]. Longer windows contain more signal and therefore improve accuracy, but they also delay the response of the hand. Their results indicated that a controller delay of roughly 100 to 125 ms gives the best balance between classification error and user performance. This value is the timing budget that any additional processing in our system must respect.

With the classifier and timing established, the next question was which features carry the most information. Phinyomark, Phukpattaranont and Limsakul compared a large number of sEMG features and showed that a small set of time-domain features provides most of the discriminative power [9]. This finding supports the use of a compact, fast feature set, which is important when sEMG is only one of several inputs in a fused system. A different direction was opened by public databases. The NinaPro database, published by Atzori et al., provided sEMG recordings of able-bodied subjects and transradial amputees performing about 50 hand and wrist movements, which made it possible to compare methods on identical data [10]. Atzori, Cognolato and Müller used this resource to apply convolutional neural networks (CNNs) to sEMG and obtained accuracies comparable to those of classical feature-based methods [11]. Deep learning is therefore a valid option for the sEMG branch, but it is not automatically superior to a well-designed LDA baseline.

Despite this methodological maturity, the clinical review by Scheme and Englehart, already introduced in Chapter 1, concluded that high offline accuracy does not guarantee usable control [4]. The classifiers described above all assume that the conditions during use are similar to those during training. The most frequent violation of this assumption in daily life is a change of limb position, which the next section examines.

4.2 The limb position effect

Fougner et al. were the first to quantify and address the limb position effect systematically [5]. Able-bodied subjects performed hand and wrist movements in several arm positions, and the classifier was trained in one position and tested in the others. The average classification error rose from about 3.8% when training and testing took place in the same position to about 18% across positions. The authors tested two remedies. The first was to train the classifier with data from several positions, which made it more general. The second was a two-stage approach in which accelerometer data first identify the current arm position and then a classifier trained for that position interprets the sEMG signal. Both remedies reduced the error, and the second introduced the idea that a separate sensor can provide the context that sEMG lacks.

Geng, Zhou and Li repeated this analysis with five transradial amputees in five arm positions [6]. This study is central to our topic, because the target users of a myoelectric prosthesis are amputees, whose residual muscles and signals differ from those of able-bodied subjects. In the amputee group, the error rose from 7.3% in the training position to 29.9% in other positions, which means that the limb position effect is even stronger for the people who need the prosthesis. Training in all five positions reduced the error to about 10.8%. The best result, about 9.0%, was obtained with a two-stage cascade in which accelerometer signals first classified the arm position, with an error of only 0.7%, and a position-specific sEMG classifier followed. The accuracy of the position estimate is important: it shows that inertial data can identify the limb position almost perfectly, so that errors in the first stage of the cascade hardly propagate to the second.

Khushaba et al. showed that the problem is not limited to the elevation of the arm [12]. Rotation of the forearm between pronation and supination, together with changes in contraction force, interacts with position and further reduces classification accuracy. A robust system must therefore consider the orientation of the forearm and not only whether the arm is raised or lowered. Taken together, these three studies define the core problem of our project in measurable terms and point to inertial sensing as the most direct known correction, which is the subject of the next section.

4.3 Inertial sensing for position awareness

If the limb position effect is caused by a change of posture, the natural correction is to measure that posture. Krasoulis et al. investigated this idea by combining sEMG with inertial measurements from accelerometers, gyroscopes and magnetometers that were integrated into the same sensor units as the electrodes [13]. In an offline study with 20 able-bodied subjects and 2 amputees performing 40 movements, adding the inertial features to the sEMG features significantly improved classification accuracy. In a subsequent real-time test with a commercial prosthetic hand, in which 11 able-bodied subjects and 1 amputee took part, the combined system also improved task completion. Good performance was reached with an average of only four to six sensor units, which shows that the additional modality does not require a complex setup.

This study differs from the cascade of Geng et al. [6] in how the inertial data are used. Krasoulis et al. simply added the inertial features to the sEMG features in one classifier, which is a form of feature-level fusion, while the cascade uses the inertial data to choose between position-specific sEMG models. Both approaches improve robustness, but they have rarely been compared on the same data. Inertial sensing has two further practical advantages: the sensors are inexpensive and already built into many commercial sEMG electrodes, and they measure exactly the physical variable that causes the limb position effect. Inertial data alone, however, say nothing about the object the user wants to grasp. To choose the correct grasp type, the system also needs information about the target, which is what computer vision can provide.

4.4 Computer vision for grasp type selection

A human chooses a grasp mainly from the shape and size of the object: a bottle invites a power grasp, a coin a precision pinch. Ghazaei et al. transferred this principle to prosthetics by training a CNN that maps an image of an object directly to one of four grasp types, namely pinch, tripod, palmar neutral and palmar pronated [14]. The network was trained on more than 500 objects, each photographed from 72 viewpoints. Offline, it classified the grasp correctly for 85% of objects seen during training and for 75% of new objects. In a test with two amputees using a webcam mounted on the prosthesis, objects were picked up successfully in up to 88% of trials. The study demonstrated that vision can propose a grasp without any muscle signal.

Earlier, Markovic et al. had followed a different route with stereovision cameras mounted on glasses [15]. Their system estimated the size and shape of the object, preshaped the prosthetic hand automatically and showed the planned grasp to the user through augmented reality feedback. The user remained in control by confirming or correcting the proposal through sEMG. This combination of automatic proposal and user confirmation is an early form of the semi-autonomous control discussed in Section 4.6.

Wang et al. moved closer to our setting by combining an object detector, RetinaNet, applied to first-person video, with sEMG classification on a public multimodal grasping dataset [16]. They found that vision reached 91.6% grasp accuracy in an early "sweet period" during the reach, when the object is clearly visible and not yet covered by the hand. Fusing the vision and sEMG decisions improved accuracy from 85.5% for sEMG alone to 90.1%. Because the vision decision was available about 800 ms before the sEMG decision, it did not add delay to the control loop. This result is important for our timing requirement from Section 4.1: a vision branch can contribute without exceeding the delay budget.

Vision has one decisive advantage for our problem: it does not depend on the posture of the arm or on the state of the muscles, so it is not affected by the limb position effect at all. Its weakness is ambiguity. In a realistic scene with several objects on a table, a camera sees all of them, and vision alone cannot decide which one the user intends to grasp. Resolving this ambiguity requires a signal of the user's intention that, like vision, is independent of the limb position. Eye-tracking gaze provides exactly such a signal.

4.5 Eye-tracking gaze for target and intent detection

Research on human motor control shows that the eyes lead the hand. Johansson et al. recorded eye and hand movements during object manipulation tasks and found that people fixate the object, and even the precise contact point, before the hand arrives there [17]. Land reviewed eye movements during everyday activities such as making tea and confirmed that gaze is directed to the next object of an action typically a fraction of a second before the hand moves towards it [18]. For a prosthesis, this behaviour is valuable for three reasons: gaze identifies the target object, it does so early, and it does not depend on the posture of the arm or on the residual muscles.

Cognolato et al. used this behaviour in a multimodal controller for a prosthetic hand [19]. A Mask R-CNN detected and segmented objects in the video of a head-mounted scene camera, and the target was defined as the last object fixated within about 250 ms before grasp onset. In parallel, a ConvLSTM network classified the grasp type from sEMG. The two branches were combined through decision-level fusion, in which only grasp types compatible with the fixated object were allowed. For amputee subjects, grasp classification accuracy rose from 63.0% to 78.6% in static tasks and from 59.0% to 74.1% in dynamic tasks, an improvement of about 15 percentage points. The gain for able-bodied subjects was smaller, about 7 percentage points, because their sEMG classification was already more accurate. The target object was identified correctly in about 80% to 92% of cases. The authors noted that the evaluation was offline and that real-world performance is likely to be lower.

This study provides the strongest evidence in the reviewed literature that context information helps most where help is needed most, namely for amputees. At the same time, it leaves our central question open: the authors did not analyse how accuracy changed with limb position, and their sEMG branch did not use inertial data. Gaze and vision tell the system *which* object is the target, while inertial data tell it *in which posture* the sEMG signal was produced. The next section reviews how such different pieces of information have been combined.

4.6 Multimodal fusion and semi-autonomous control

Markovic et al. built one of the first prostheses that combined several modalities in a working system, which they called a context- and user-aware prosthesis (CASP) [20]. Computer vision estimated the properties of the target object, inertial sensing tracked the orientation of the prosthesis, and sEMG gave the user supervisory control. The system preshaped the hand and oriented the wrist automatically, while the user decided when to open, close or correct it. With all degrees of freedom active, CASP performed significantly better than a conventional myoelectric controller in a study with 10 able-bodied subjects and 1 amputee, and the participants made fewer compensatory movements during reaching. Mouchoux et al. later extended this semi-autonomous approach and tested it with both amputee and able-bodied participants [21]. Artificial perception increased task performance and reduced the effort reported by users compared with conventional myoelectric control.

Across the reviewed studies, three levels of fusion can be distinguished, summarised in Table 2. In feature-level fusion, the features of all sensors are joined into one input vector for a single classifier, as in the sEMG and inertial system of Krasoulis et al. [13]. In hierarchical or cascade fusion, one modality selects the model used for another, as in the position-specific classifiers of Fougner et al. [5] and Geng et al. [6]. In decision-level fusion, each modality produces its own decision or probability and a rule or probabilistic step combines them, as in the gaze and vision systems of Cognolato et al. [19] and Wang et al. [16].

Table 2: Fusion levels found in the reviewed literature.

Fusion level	Principle	Example in the literature	Strength	Weakness
Feature level	Features of all sensors joined into one classifier	sEMG and inertial features [13]	Learns interactions between modalities	Needs synchronised data; one faulty sensor affects the whole decision

Fusion level	Principle	Example in the literature	Strength	Weakness
Hierarchical (cascade)	One modality selects the model of another	Limb position selects the sEMG classifier [5], [6]	Simple; directly targets the limb position effect	Errors of the first stage propagate
Decision level	Each modality decides; a rule or probabilistic step combines the decisions	Gaze, vision and sEMG [16], [19]	Modular; can fall back when a modality is missing	Cannot learn low-level interactions

No single level is best for all modalities. The cascade suits inertial data, because the limb position is a discrete context that changes the meaning of the sEMG signal. Decision-level fusion suits gaze and vision, because they produce an independent opinion about the target object that can be checked against the sEMG decision, and because the system can continue to work if the user looks away or the object is hidden. This observation is the basis of the hybrid architecture proposed in Chapter 6.

4.7 Public datasets and evaluation metrics

A Software and Simulation project depends on open data that contain all the required signals, recorded synchronously and from the target user group. Two public datasets were considered. The NinaPro database [10], introduced in Section 4.1, is the standard benchmark for sEMG classification. It contains several sub-databases with able-bodied subjects and transradial amputees performing about 50 repeated hand and wrist movements, recorded with sEMG electrodes, a data glove and, in the later sub-databases, accelerometers. It is well suited for developing and benchmarking the sEMG baseline, but it does not include gaze or scene video.

The MeganePro dataset (MDS1), published by Cognolato et al., was recorded specifically for multimodal prosthetics research and is the dataset used by the gaze and vision studies reviewed above [16], [19], [22]. It contains data from 15 transradial amputees and 30 able-bodied subjects. sEMG was recorded with 12 Delsys Trigno electrodes at 1926 Hz, each with an integrated 3-axis accelerometer sampled at 148 Hz. A Tobii Pro Glasses 2 eye tracker recorded gaze at 100 Hz, scene video at 25 frames per second and head motion. The participants performed 10 grasp types on 18 household objects in static and dynamic tasks, both seated and standing. Because the accelerometers are located on the forearm,

the posture of the limb can be estimated for every trial. MeganePro therefore contains all four modalities of our topic, synchronised, and includes amputee subjects, which makes it the main dataset of the project. Table 3 compares the two datasets.

Table 3: Public datasets considered for the project.

Dataset	Subjects	Sensors	Tasks	Role in our project
NinaPro [10]	Able-bodied subjects and transradial amputees, several sub-databases	sEMG; data glove; accelerometers in later sub-databases	About 50 hand and wrist movements, repeated	Benchmark for the sEMG baseline
MeganePro MDS1 [22]	15 transradial amputees, 30 able-bodied	12 sEMG channels (1926 Hz) with forearm accelerometers (148 Hz); gaze (100 Hz); scene video (25 fps); head motion	10 grasp types on 18 objects; static and dynamic; seated and standing	Main dataset: all four modalities, synchronised

The grasp classes in such datasets are based on established taxonomies of human grasping. Feix et al. organised human grasp types into the GRASP taxonomy according to the opposition type, the virtual fingers involved and the power or precision character of the grasp [23]. Using a taxonomy-based class set makes our results comparable with other work and allows a meaningful mapping from objects to compatible grasp types in the vision branch.

The reviewed studies use a consistent set of evaluation metrics, which we adopt. The primary metric is offline classification accuracy, or its complement the error rate, reported per subject and averaged separately for amputees and able-bodied subjects. To measure robustness, the inter-position error is used: the classifier is trained in one limb position and tested in another, as in Fougner et al. [5] and Geng et al. [6]. Timing is measured as decision latency relative to grasp onset, with the delay budget of about 100 to 125 ms identified by Smith et al. [8]. For the gaze branch, the rate of correct target-object identification is reported [19]. Finally, real-time functional tests such as the Target Achievement Control test of Simon et al. [24] evaluate a controller with a user in the loop. Such a test is outside the scope of a purely offline simulation, but it describes the natural next step after our project.

5 Synthesis: From the Literature to Our Topic

5.1 Comparison of the reviewed works

Table 4 places the main experimental studies side by side, in the order in which they appear in Chapter 4. Read from top to bottom, the table shows a clear progression: from sEMG alone, to sEMG with inertial data, to vision, gaze and their fusion. Each row also shows the limitation that motivated the next step.

Table 4: Comparison of the main experimental studies.

Ref.	Modalities	Method	Key result	Limitation for our problem
[3], [7]	sEMG	Time-domain features with LDA, real time	Multi-class myoelectric control	Assumes a fixed limb position
[5]	sEMG, accelerometer	Multi-position training; two-stage classifier	Error 3.8% to 18% across positions; reduced by position-aware methods	Able-bodied subjects only
[6]	sEMG, accelerometer	Two-stage cascade	Amputees: error 29.9% reduced to about 9.0%	Five positions only; LDA only; no object information
[12]	sEMG	Analysis of forearm orientation and force	Orientation and force further reduce accuracy	Describes the problem; no multimodal solution
[13]	sEMG, inertial	Feature-level fusion	Significant offline and real-time gains with 4 to 6 sensors	No information about the target object
[14]	Vision	CNN from object image to grasp type	85% on seen, 75% on new objects	Cannot tell which object is the target
[16]	Vision, sEMG	RetinaNet with decision fusion	sEMG 85.5% to fused 90.1%	No gaze; limb position not analysed

Ref.	Modalities	Method	Key result	Limitation for our problem
[19]	Gaze, vision, sEMG	Mask R-CNN and ConvLSTM, decision fusion	Amputees: about +15 percentage points	Offline; no inertial data; limb position not analysed
[20], [21]	Vision, inertial, sEMG	Semi-autonomous control	Better performance, less effort than conventional control	Custom hardware; small samples; not reproducible offline

5.2 Identified research gaps

The comparison reveals four research gaps. The first and most important is that no reviewed study fuses all four modalities while testing robustness to limb position. The studies on the limb position effect [5], [6], [12] use only sEMG and accelerometers and do not consider the target object, while the studies that use gaze and vision [16], [19] report accuracy without separating it by limb position. It is therefore unknown how much of the remaining inter-position error can be removed when inertial context and target-object context are combined.

The second gap concerns the use of inertial data. In the reviewed work, inertial information is used either as an additional feature [13] or as a selector of position-specific models [5], [6], but the two strategies have rarely been compared on the same data and the same subjects. It is therefore not clear which strategy is better suited for a fused system that also includes gaze and vision.

The third gap is graceful degradation. In daily life, users look away from the object, objects are hidden behind others, and sensors lose contact or fail. A fused controller must continue to work in these situations, at least as well as sEMG alone. However, the behaviour of a fused system when one modality is missing is seldom evaluated in the reviewed studies.

The fourth gap is reproducibility. Several of the systems with the strongest results [20], [21] use custom hardware and data that are not publicly available. Open, documented code that runs on a public multimodal dataset and regenerates the reported results is rare, which makes it difficult to compare methods fairly.

5.3 How each gap shaped the topic

The four gaps were not only identified after the review; they directly determined the wording of our topic and the design of our system. Figure 1 traces this path from the clinical problem through the literature to each part of the title.

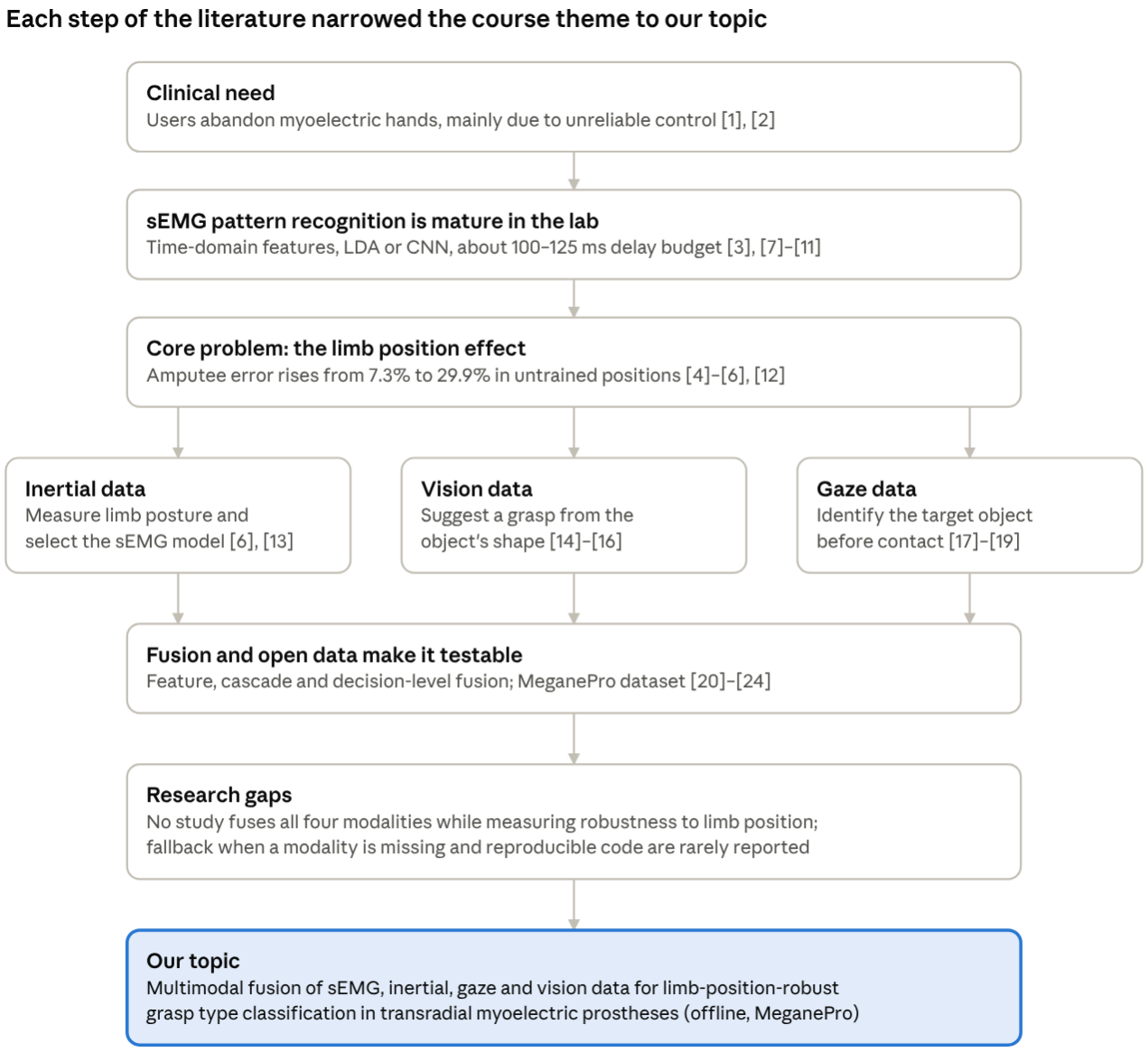


Figure 1: literature map · from the clinical need through the reviewed works to the project topic

Reading Figure 1 from top to bottom, every step narrows the previous one, and every box rests on the references cited in it.

The first gap gives the topic its core: *multimodal fusion of sEMG, inertial, gaze and vision data* for *limb-position-robust* classification. The quantitative results of Fougner et al. [5] and Geng et al. [6] provide the robustness criterion, and the gains reported by Cognolato et al. [19] provide a realistic target for the improvement. The second gap determines how the inertial data are used: our system will compare feature-level fusion with a cascade on the same data, instead of assuming one strategy. The third gap explains why decision-level fusion was chosen for gaze and vision, since it allows the system to fall back to sEMG and inertial data when the target cannot be identified, and why this fallback is explicitly evaluated. The fourth gap determines the project track and the evaluation setting: an offline Software and Simulation study on the public MeganePro dataset [22], with all code, data splits and results on GitHub. Finally, the user group and the output follow from

the clinical literature and the dataset: *transradial* users, because they retain forearm muscles for sEMG and are represented in MeganePro, and *grasp type classification*, because selecting the correct grasp before the hand closes is exactly the decision that the limb position effect disturbs.

6 Proposed System and Method Outline

6.1 System definition

Based on the research gaps, our project develops a hybrid fusion pipeline that classifies the intended grasp type of a transradial amputee and remains accurate across limb positions. The pipeline is developed and evaluated offline on the MeganePro dataset [22], and all code and documentation are kept in the project's GitHub repository. Figure 2 shows the structure of the pipeline, and Table 5 defines its inputs, processing, output and users.

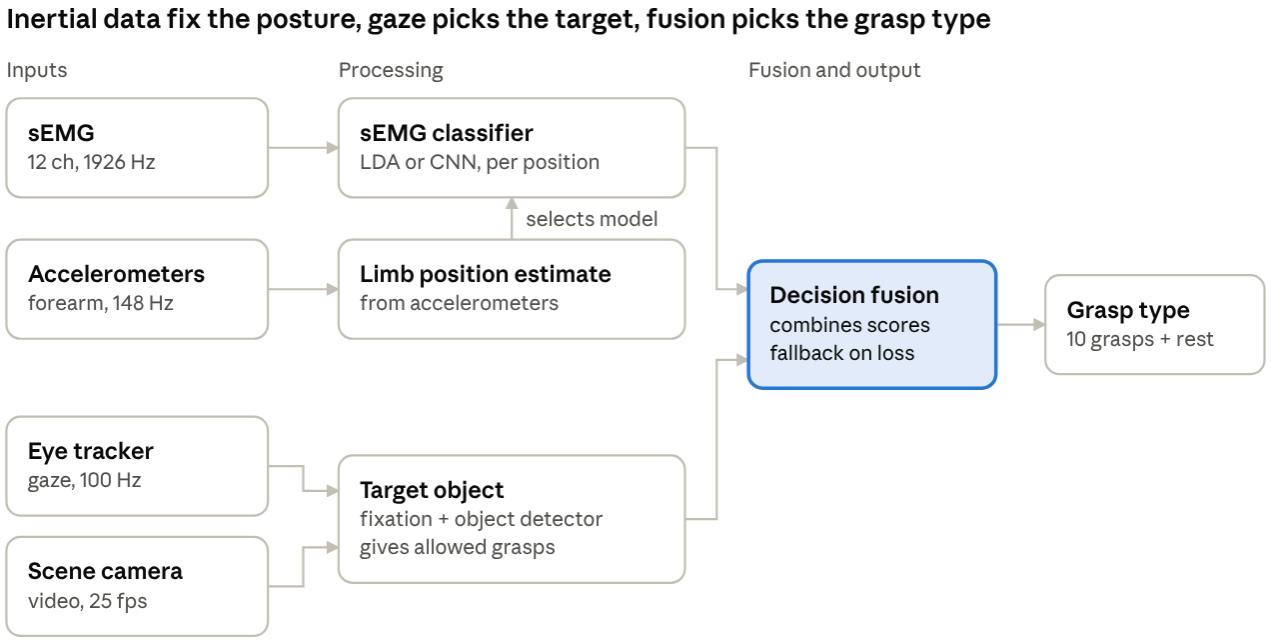


Figure 2: proposed fusion pipeline · 4 inputs, 3 processing branches, 1 fusion step

Table 5: Definition of the proposed system.

Element	Specification
Inputs	sEMG (12 channels, 1926 Hz); forearm accelerometers (148 Hz); gaze point (100 Hz); scene video (25 fps)

Element	Specification
System	Pre-processing and synchronisation; limb position estimation from inertial data; position-aware sEMG classifier; gaze-based target object detection in the scene video; mapping from object to compatible grasp types; decision-level fusion with fallback
Output	One of 10 grasp types, or rest, with a confidence value, issued before grasp onset
Target users	Transradial amputees using multi-grasp myoelectric hands

The inertial branch addresses the limb position effect directly. It estimates the limb position from the orientation of the forearm accelerometers and selects, or informs, the sEMG classifier accordingly, following the cascade principle of Geng et al. [6]. The gaze and vision branch addresses the ambiguity of the grasp choice. It detects the objects in the scene video, identifies the fixated object shortly before grasp onset, as proposed by Cognolato et al. [19], and converts it into a set of allowed grasp types. The fusion step combines the probability outputs of both branches into one decision. If gaze or vision is not available, for example because the user looks away, the system falls back to the sEMG and inertial branch instead of failing.

6.2 Requirements and target values

Each requirement is derived from a result in the literature, so that the targets are realistic and verifiable. The first requirement concerns accuracy: grasp type classification accuracy for amputee subjects, averaged across limb positions, should exceed the sEMG-only baseline by at least 10 percentage points, which is below the gain of about 15 points reported for gaze and vision fusion [19] and therefore achievable. The second requirement concerns robustness: the drop in accuracy between the training position and other limb positions should be at most half of the drop measured for the sEMG-only baseline, in line with the reduction achieved by the cascade of Geng et al. [6]. The third requirement concerns timing: the processing added by fusion should not exceed about 125 ms per decision, the upper end of the delay budget identified by Smith et al. [8]. The fourth requirement concerns safety of operation: when gaze or vision data are missing, the system must still output a decision whose accuracy is not lower than that of the sEMG and inertial branch alone. The fifth requirement concerns reproducibility: data splits must be fixed and documented, and all reported results must be regenerated from the repository by one script.

6.3 Planned method and evaluation

The work is organised in six steps that follow the structure of the review and the course milestones. In the first step, the MeganePro signals are loaded, synchronised and segmented into analysis windows aligned with grasp onset. In the second step, an sEMG-only baseline is built with Hudgins features and LDA [3], [7], and compared with a small CNN [11]. In the third step, limb position labels are derived from the orientation of the forearm accelerometers, and the limb position effect is measured on the baseline by training and testing in different positions. In the fourth step, inertial data are added in two ways, as additional features and as a cascade that selects position-specific classifiers, and both are compared on the same splits. In the fifth step, the fixated object is identified with a pretrained object detector and gaze fixations, and it is mapped to a prior over compatible grasp types based on the GRASP taxonomy [23]. In the sixth step, all branches are fused at decision level and the complete system is evaluated per limb position, with each modality removed in turn (ablation) and with the added latency measured.

Results will be reported separately for amputee and able-bodied subjects, because the literature shows that the benefit of context information differs strongly between the two groups [19]. The baseline and the measured limb position effect form the methodology milestone of 18 November; the fused system and its ablation form the experiments and demonstration milestones in December.

6.4 Scope and limitations

The project is an offline simulation. It does not include a physical prosthesis, a real-time experiment with users or a functional test such as the Target Achievement Control test [24], so the results will indicate potential rather than prove clinical benefit. The evaluation is limited to the subjects, objects, grasp types and limb positions contained in MeganePro, and the limb position classes must be derived from accelerometer orientation because they are not labelled directly in the dataset. Invasive interfaces, sensory feedback and continuous force control remain outside the scope, as defined in Chapter 2.

7 Conclusion

This review followed a single line of reasoning from a clinical need to a specific, testable project topic. The clinical literature shows that many people with upper-limb loss abandon their myoelectric prosthesis, and that unreliable control is one of the main reasons [1], [2]. Myoelectric pattern recognition is a mature method with well-established features, classifiers and timing limits [3], [7], [8], [9], but its accuracy depends on conditions

remaining the same as during training [4]. The most frequent and best-documented violation of this assumption is a change of limb position, which raises the classification error of transradial amputees from 7.3% to 29.9% [6].

The review then showed that each additional modality corrects a different part of this problem. Inertial data measure the posture of the limb and make position-aware classification possible, which reduced the inter-position error of amputees to about 9% [5], [6], [13]. Computer vision proposes a grasp type from the shape of the object, independent of the muscle signals and therefore unaffected by the limb position [14], [16]. Gaze reveals which object the user intends to grasp before the hand reaches it [17], [18], and the fusion of gaze, vision and sEMG improved the grasp classification accuracy of amputees by about 15 percentage points [19]. Semi-autonomous systems that combine perception with user control increased performance and reduced effort [20], [21].

The synthesis identified four gaps: no study fuses sEMG, inertial, gaze and vision data while measuring robustness to limb position; the two ways of using inertial data have rarely been compared; graceful degradation when a modality is missing is seldom evaluated; and reproducible implementations on public data are rare. Each gap shaped one part of our topic, *Multimodal Fusion of sEMG, Inertial, Gaze and Vision Data for Limb-Position-Robust Grasp Type Classification in Transradial Myoelectric Prostheses*, and the public MeganePro dataset [22] makes it possible to address all four within one semester. The next milestone will turn the method outline into a detailed methodology, implement the sEMG baseline and measure the limb position effect on it, which provides the reference against which every later improvement will be judged.

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